

Phase 1: Feature Engineering & Baseline Signature Distribution



A clear, intuitive reference explaining **Phase 1** of Question 5: How we take thousands of raw tumour genes and turn them into neat, easy-to-understand immune scorecards.

1. The High-Dimensional RNA-Seq Challenge & Feature Engineering

Analytical Methodology & Rationale

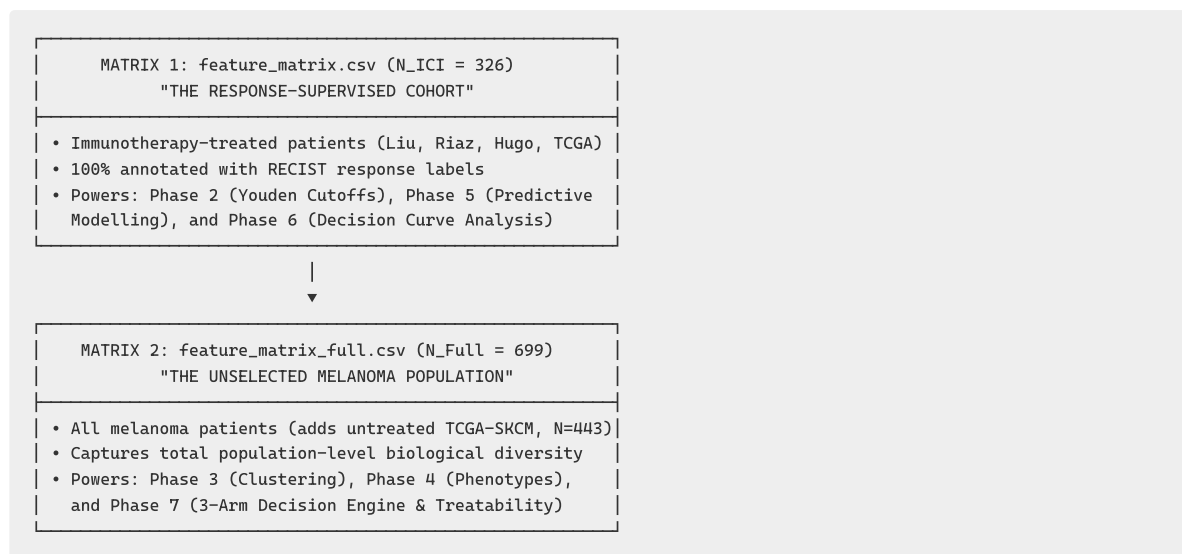
- **What is being done:** Loading multi-modal clinical, transcriptomic, and genomic datasets to engineer 17 biologically interpretable features across dual cohort feature matrices ($N_{ICI} = 326$ and $N_{Full} = 699$).
- **Why we are doing it:** Raw bulk RNA-seq gene expression matrices contain ~19,757 genes per sample. High-dimensional feature spaces ($p \gg n$) cause severe overfitting, collinearity, and loss of biological interpretability. Projecting expression profiles onto validated immune signature scores and cell deconvolution fractions compresses noise into actionable clinical signals.
- **What question it answers:** How do we clean, harmonise, and condense thousands of complex tumour genes into structured biological feature matrices that downstream response prediction models and patient clustering algorithms can learn from?

The Dimensionality Reduction Strategy

- **The $p \gg n$ Problem:** Raw sequencing measures ~19,757 protein-coding genes per tumour sample across $N = 699$ patients. Feeding raw gene matrices directly into machine learning algorithms creates severe curse-of-dimensionality issues and redundant feature weights.
- **The Biological Solution:** Instead of evaluating unconstrained individual genes, we project gene expression onto **17 biologically validated axes** (6 immune signatures, 4 macrophage polarisation scores, 7 cell deconvolution fractions) and **9 genomic/TMB markers**.
- **Translational Relevance:** Clinicians cannot interpret arbitrary weights across thousands of genes, but they can easily interpret established biomarkers like Tumour Inflammation Signature (**TIS**), Cytolytic Index (**CYT**), or **CD274** (**PD-L1**) expression.

2. Dual Feature Matrix Dataflow Architecture

To support distinct machine learning requirements across downstream analytical phases, Phase 1 constructs two separate feature matrices:



Why Two Datasets Are Necessary

1. **Supervised Model Training** ($N_{ICI} = 326$):
 - To fit predictive classifiers or determine diagnostic sensitivity/specificity cutoffs, algorithms require ground-truth treatment outcome labels (**Responder** vs **Non-Responder**).
 - `feature_matrix.csv` strictly isolates patients who received anti-PD-1 or anti-CTLA-4 immunotherapy across 4 trial cohorts (**Liu 2019**, **Riaz 2017**, **Hugo 2016**, and ICI-treated **TCGA-SKCM**).
2. **Unsupervised Stratification & Real-World Simulation** ($N_{Full} = 699$):
 - In clinical practice, new primary or surgical patients arrive before receiving systemic therapy.
 - `feature_matrix_full.csv` merges trial cohorts with the complete reference cohort (**TCGA-SKCM**, $N = 443$), capturing population-wide biological spectrums without biasing unsupervised clustering toward pre-treated trial populations.

3. Microenvironment Deconvolution: Cops, Soldiers & Barriers

Tumour tissues are heterogeneous microenvironments composed of malignant cells, infiltrating immune cells, and dense stromal barriers. We can understand the microenvironment by looking at four key biological roles:

-  **M1 Macrophages ("Good Cops" — Antitumour Inflammation):**
 - Express pro-inflammatory markers (`TNF` , `IL12B` , `CXCL10` , `NOS2` , `IRF5`). They act like vigilant police officers that actively swallow tumour debris and present antigens, sounding the alarm to recruit cytotoxic T cells.
-  **M2 Macrophages ("Double Agents" — Immunosuppressive Shielding):**
 - Express anti-inflammatory markers (`CD163` , `MRC1` , `MSR1` , `TGFB1` , `ARG1`). Rather than attacking cancer, they act like corrupt double agents that secretively release immunosuppressive cytokines, promote tumour blood vessel growth (angiogenesis), and disarm infiltrating T cells.
-  **CD8+ Cytotoxic T Cells ("T-Cell Soldiers" — Frontline Effector Cells):**
 - Measured by `CD8A` , `CD8B` , `CD3D` , and `CD3E` . These are the elite frontline soldiers capable of destroying malignant cells. Anti-PD-1 (`CD274`) immunotherapy works by unbinding checkpoint breaks on these soldiers so they can release cytolytic weapons (Perforin `PRF1` and Granzyme A `GZMA`).
-  **Cancer-Associated Fibroblasts (`CAFs` , "Scar Tissue Physical Barriers"):**
 - Express stromal markers (`FAP` , `PDGFRB` , `COL1A1` , `ACTA2`). They deposit dense collagen scar tissue around the tumour, creating a physical fortress wall that prevents T-cell soldiers from penetrating into the tumour core (immune exclusion).

How Transcriptomic Cell Deconvolution Was Determined

Think of bulk tumour RNA sequencing like taking a complex fruit smoothie and figuring out how much apple, banana, and strawberry went into it!

Phase 1 determines cell type abundances using two complementary approaches in `deconvolution.py`:

1. **The Average Marker Scorecard Method (`compute_cell_deconvolution`):**
 - For each cell type (e.g. CD8+ T cells or CAFs), we select a panel of 3–5 specific marker genes that only that cell type produces in high amounts.
 - We calculate each patient's score by taking the average expression level across those marker genes.
 - **Key Marker Panels:** `CD8_T_cells` (`CD8A` , `CD8B`), `NK_cells` (`NCAM1`), `B_cells` (`CD19`), `M1_Macrophages` (`TNF`), `M2_Macrophages` (`CD163`), and `CAFs` (`FAP` , `PDGFRB` , `COL1A1`).
2. **The Macrophage Polarisation Scorecard (`compute_macrophage_stv`):**
 - Rather than checking only 3 markers, we compare patient expression against a 14,837-gene scorecard (`m1_m2_stv.csv`).
 - Genes with positive weights build the **M1 Score** ("Good Cop"), while negative weights build the **M2 Score** ("Double Agent").
 - Combining these produces the **M1/M2 Ratio** (ranging from 0.0 to 1.0), showing whether the tumour microenvironment is dominated by antitumour inflammation or immunosuppressive stromal resistance.
 - **High M1/M2 Ratio:** M1-dominated, inflamed microenvironment highly responsive to anti-PD-1 monotherapy.
 - **Low M1/M2 Ratio:** M2-dominated stromal suppression. Even if total T-cell density is moderate, M2 dominance disarms T-cell soldiers, signaling a requirement for combination helper therapies.

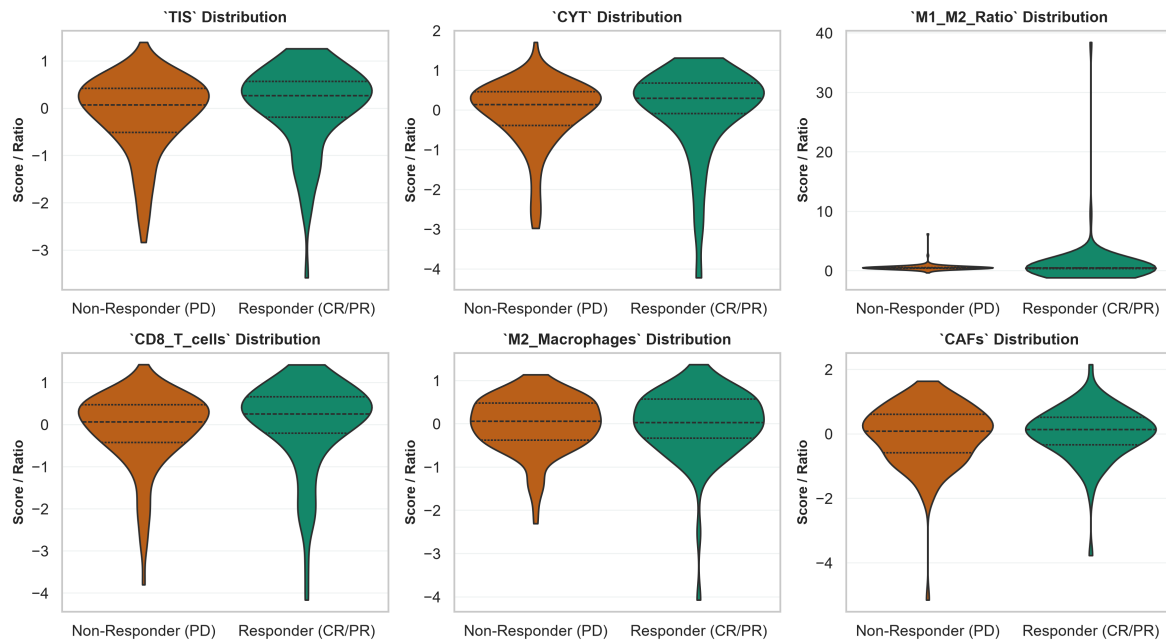
4. Complete Feature Inventory Breakdown

Phase 1 condenses ~19,757 genes into 17 transcriptomic features and 9 genomic/TMB features:

Feature Category	Features Included	Primary Biological Signalling / Function
Core Immune Signatures (6)	<code>TIS</code> , <code>CYT</code> , <code>IFN_gamma</code> , <code>CD8_Tcell</code> , <code>IMPRES</code> , <code>PD_L1</code> (<code>CD274</code>)	Quantifies active IFN- γ signalling, cytotoxic killing capability (<code>PRF1</code> , <code>GZMA</code>), and checkpoint target expression.
Macrophage STV (4)	<code>M1_score</code> , <code>M2_score</code> , <code>M1_M2_Ratio</code> , <code>Macrophage_STV_Score</code>	Resolves functional macrophage polarisation using a 14,837-gene linear vector.
Cell Deconvolution (7)	<code>CD8_T_cells</code> , <code>CD4_T_cells</code> , <code>NK_cells</code> , <code>B_cells</code> , <code>M1_Macrophages</code> , <code>M2_Macrophages</code> , <code>CAFs</code>	Marker-based relative cell abundance estimates distinguishing immune infiltrates from stromal fibroblasts.
Genomic & Neoantigens (9)	<code>mut_BRAF</code> , <code>mut_NRAS</code> , <code>mut_NF1</code> , <code>TMB_NONSYNONYMOUS</code> , Neoantigen counts	Identifies actionable oncogenic drivers (MAPK pathway activation) and mutational neoantigen load.

5. Visualising Baseline Feature Distributions

Baseline Immune & Microenvironmental Biomarker Distributions (N=195)



Baseline Biomarker Observations

- Immune Hot Characteristics:** Immunotherapy Responders (CR/PR) exhibit significantly higher baseline levels of TIS, CYT, IFN_gamma, and CD8_Tcell than Non-Responders (PD).
- Stromal Exclusion Mechanisms:** Non-responders frequently present with elevated CAFs (scar tissue walls) or low M1_M2_Ratio ("Double Agent" dominance), demonstrating that high stromal density or M2 macrophage polarization suppresses immune efficacy even in tumours with modest T-cell presence.

Key Takeaways

- Feature Compression:** Transformed raw high-dimensional transcriptomics (~19,757 genes) into 17 interpretable biological signatures across 26 total clinical, genomic, and microenvironmental features.
- Dual Cohort Architecture:** Standardized `feature_matrix.csv` ($N_{ICI} = 326$) for response prediction models and `feature_matrix_full.csv` ($N_{Full} = 699$) for unsupervised stratification and 3-arm decision tree routing.
- Microenvironment Profiling:** Captured stromal immune exclusion via the 14,837-gene Macrophage STV score, providing a quantitative basis for identifying combination-therapy candidates.

Key Phase 1 Limitations to Keep in Mind

- Relative Scores vs True Percentages:** Deconvolution uses mean marker log-expression (`compute_cell_deconvolution`), producing relative arbitrary scores rather than true constrained 0–100% cell percentage proportions (like CIBERSORTx).
- Spatial Blindness:** Bulk RNA-seq blends the whole biopsy together. It cannot tell if T-cell soldiers are physically inside the tumour nest (*Immune Hot*) or trapped outside in the scar tissue wall (*Immune Excluded*).
- Indirect DNA Proxies:** Because trial cohorts lack Copy Number Alteration (CNA) GISTIC files, Phase 1 relies on Tumour Mutational Burden (`TMB_NONSYNONYMOUS`) and single-gene transcript levels (`PTEN`, `CDKN2A`) as indirect proxies for chromosomal instability.
- 1D Macrophage Axis:** The `M1_M2_Ratio` models macrophages as a single "Good Cop vs Double Agent" spectrum, whereas real macrophages exist in multi-dimensional states (e.g., M2a, M2c, lipid-laden).
- Static Snapshot:** Pre-treatment biopsies cannot capture early dynamic immune changes occurring 2–4 weeks after starting anti-PD-1 (`CD274`) treatment.